Interaction Insights

Shareable brief. The source is the GA4 backup in gs://goods-ga4-backup_analysis, shards events-000000000000.parquet through events-000000000049.parquet. Reaction rates, page time, journeys, session gaps, and power users are read from BigQuery after those shards are loaded.

Four questions

1. How do users react?

BigQuery view v_reaction_rates. Click rate is click and ad_click divided by page views. Ignore rate is ad_ignore divided by page views. Conversion rate is purchase, sign_up, subscribe, and ad_free_subscribe divided by sessions. ad_close and app_reopen are counted on their own.

2. Time on a page, and the journey

v_page_time sums engagement_time_msec on a page and divides by page views. v_user_journey and v_journey_patterns show the first four steps of a session. A page_view is labeled with its page path. Seconds between steps use the export timestamps.

3. What causes a session to end?

v_session_end_proxy is the last event in each session. v_session_breaks is a gap of 30 minutes or more between a user's events. Path exploration and retention in the GA4 interface add the why.

4. How do power users use the product?

v_power_users keeps a user with 15 or more events, or 3 or more sessions, or 8 or more page views, or a conversion. power_score = events + 10 x sessions + 20 x conversions.

How the warehouse is filled

scripts/download_ga4.sh copies the 50 parquet shards into data/ with gcloud storage cp. The bucket is private. setup_bq.py creates dataset interaction_insights, table ga4_events, and the views. replay.py appends events with BigQuery load jobs about every 60 seconds. No streaming inserts and no DML. Views keep the latest replay_run_id. The table expires about 60 days after it is created. The loader does not send the backup through the Measurement Protocol.

Open it

python live_app.py --port 8765 --speed 80 --max-rows 1000
http://127.0.0.1:8765/
With the shards in data/, the page replays that export. Without them it shows a labeled stand-in and says so. With GCP_PROJECT and the service account set, the page can read the same views from BigQuery. Never commit the parquet, the JSON key, or a filled-in .env.